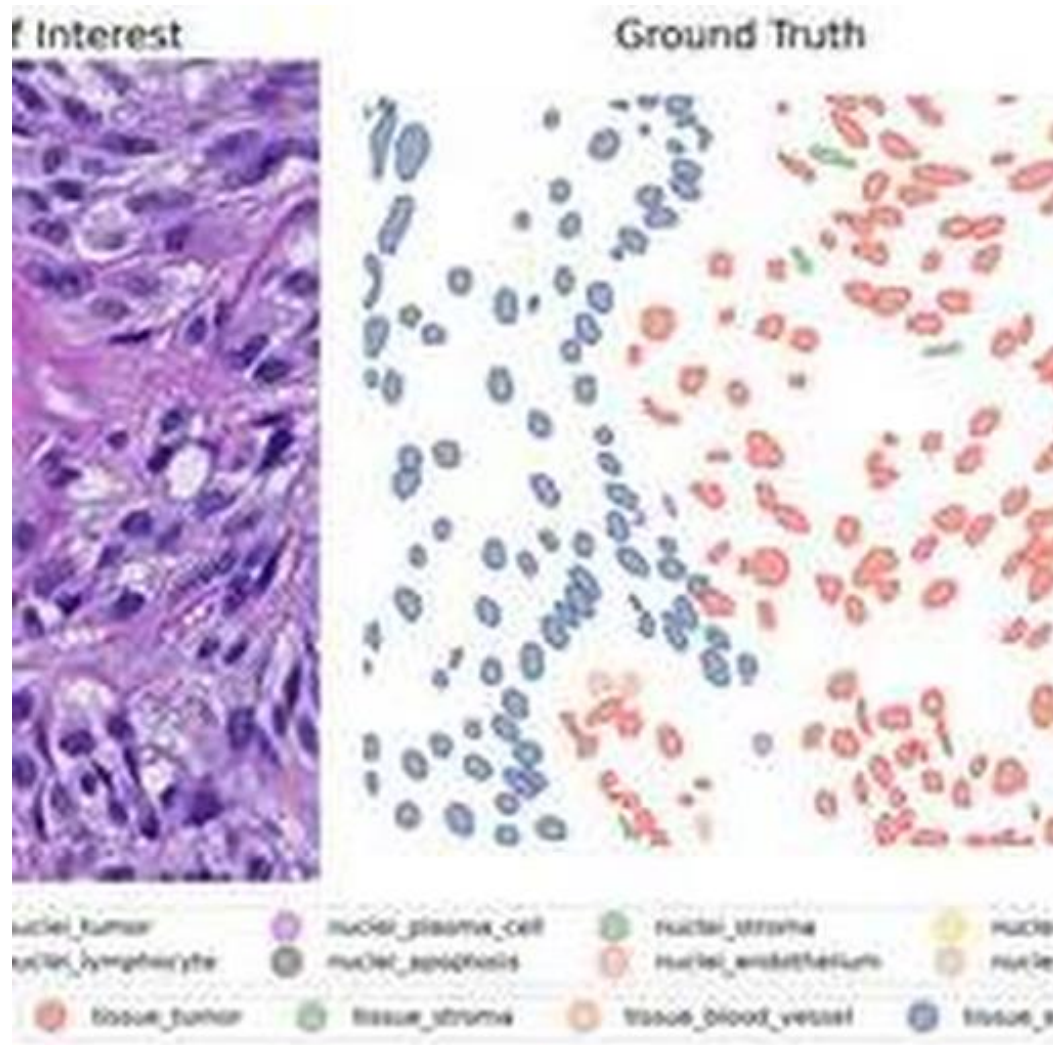


5. Nuclei Instance Segmentation Across Tissue Types



Source: PanNuke / related dataset visual. <https://jgamper.github.io/PaNukeDataset/>

5. Nuclei Instance Segmentation Across Tissue Types

Domain: Digital pathology; histology; instance segmentation; cell classification; robustness.

Dataset: Use PanNuke as the primary dataset: about 7,900 H&E patches from 19 tissue types with about 190,000–200,000 annotated nuclei in five classes.

Core question: Can one nuclei instance-segmentation / classification model perform consistently across diverse tissue types and nucleus classes?

Why this is interesting: Touching nuclei must be separated as individual objects, making the task more demanding than semantic segmentation and allowing tissue- and class-specific analysis.

Webpage / References: <https://jgamper.github.io/PanNukeDataset/> ;
<https://monuseg.grand-challenge.org/>

5. Nuclei Instance Segmentation Across Tissue Types

— Student Task

Task:

1. Reproduce a strong instance-segmentation/classification model such as HoVer-Net on PanNuke.
2. Evaluate by tissue type and nucleus class using instance-level metrics such as Panoptic Quality or AJI.
3. Analyze merged/split/missed nuclei and test one targeted improvement for difficult tissues or rare classes.

Difficulty: Medium-high.

Computational requirements: About 8–16 GB GPU memory for HoVer-Net; 16–24 GB for optional transformer-based models.

Webpage / References: <https://jgamper.github.io/PanNukeDataset/> ;
<https://monuseg.grand-challenge.org/>

Summary

1. For every project, require two stages: (a) establish a reproducible baseline, and (b) develop, modify, or substantially adapt at least one method.
2. Project goal is not only to compare existing algorithms, but also to build or improve something.
3. The current seven-project set provides a good methodological spread: algorithms / optimization, image processing, ML / statistics, domain generalization, instance segmentation, multimodal learning, and foundation-model transfer learning.

5. Nuclei Instance Segmentation

Simple explanation: In pathology images, we want to find every individual cell nucleus and separate touching nuclei from each other. This is harder than ordinary segmentation because two nuclei that touch must still be identified as two separate objects. Students can also study how performance changes across different tissue types and nucleus classes.

Quick points to remember:

- **Input:** small H&E pathology image patches
- **Output:** a separate mask / identity for every nucleus
- **Dataset:** PanNuke: about 7,900 patches, 19 tissue types, about 205,000 nuclei
- **Simple baseline:** HoVer-Net or another established instance-segmentation model
- **Hard parts:** touching nuclei, small nuclei, tissue variability
- **Possible improvement:** reduce merged, split, or missed nuclei
- **Main question:** How can we detect and separate individual nuclei more accurately?